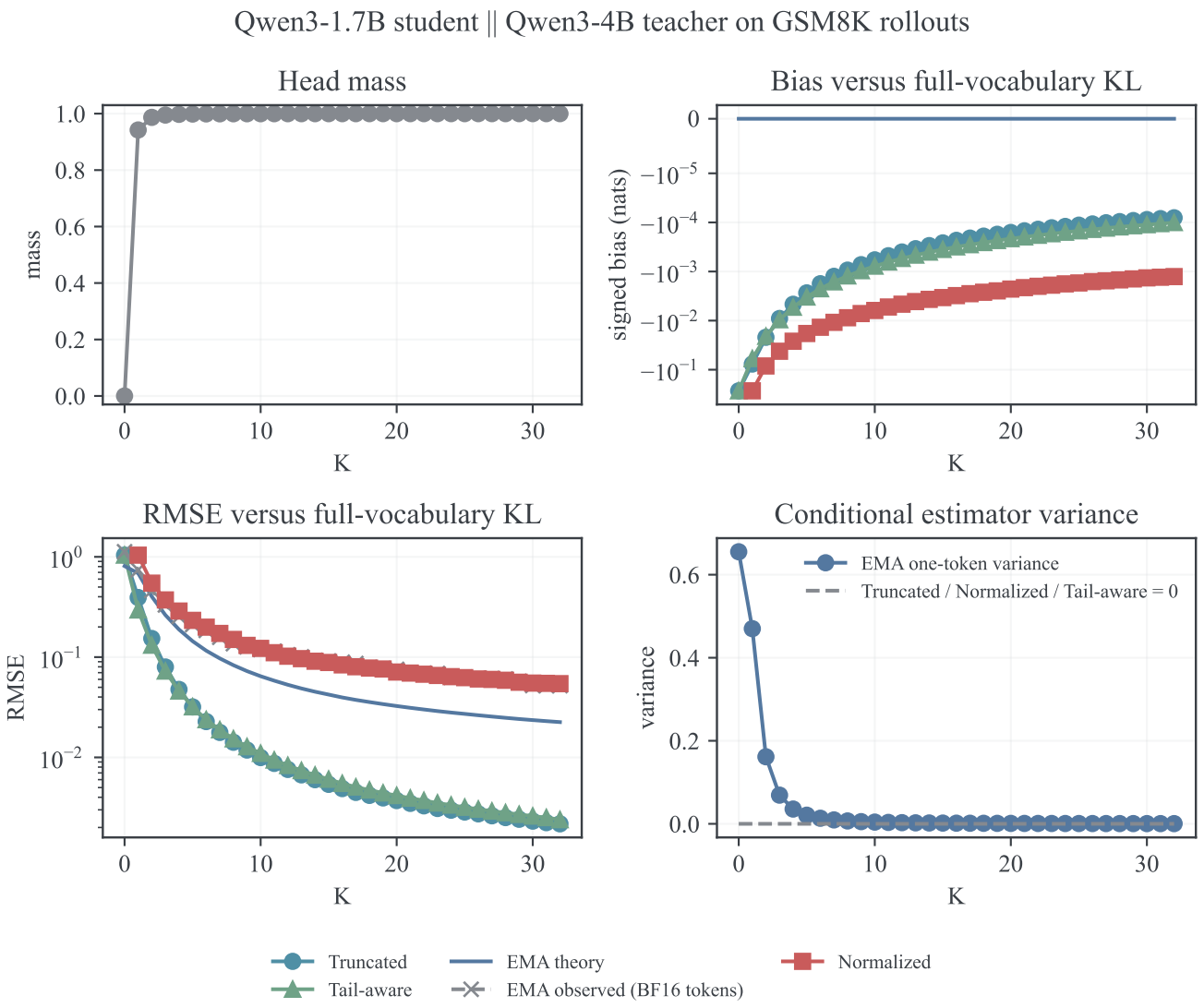
Qwen3-1.7B student || Qwen3-4B teacher on GSM8K rollouts
Head mass
mass
K
Bias versus full-vocabulary KL
signed bias (nats)
K
RMSE versus full-vocabulary KL
RMSE
K
Conditional estimator variance
EMA one-token variance
Truncated / Normalized / Tail-aware = 0
variance
K
Truncated
Tail-aware
EMA theory
EMA observed (BF16 tokens)
Normalized